

PLASMA SPRAY PROCESSORS

WORK INSTRUCTION

Quality Assurance and Calibration

Document No.	WI-QMS-13	Sections	7 (see contents)
Title	Quality Assurance and Calibration		
Process Owner	DH Quality	Compiled	14-Aug-2026
ISO 9001:2015 Clauses	6.2, 7.1.5, 7.1.5.1, 7.1.5.2, 8.2, 8.2.1, 8.2.2, 8.4, 8.4.1, 8.4.2, 8.5.1, 8.5.2, 8.6, 8.7, 9.1, 9.1.3, 10.2		

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Note on This Consolidation

This Work Instruction consolidates all Quality Assurance and Calibration documentation (QMS-13) into a single reference: the core QA procedure and its 2024 revision, the calibration procedure, the QA turtle diagram, the QA process-flow diagram, the NDT visual-testing standard with its reference defect photographs, and the current Quality Control Plan.

The two source workbooks supplied for this consolidation (the original and the file named “REVISE”) contained identical data — the same text and the same embedded images throughout — so there was no conflicting content to reconcile. The section that follows is the complete, unified set.

Original layouts, tables, diagrams and reference photographs are reproduced as formatted in the source workbook to preserve their visual accuracy, particularly for the turtle diagram, process-flow diagram and NDT defect-reference images.

Plasma Spray Processors

Title: Quality assurance and calibration:

Ver. 00 / 01.04.2019 QMS-13

ISO Clause: 6.2,7.1.5,7.1.5.1,7.1.5.2,8.2,8.2.1,8.2.2,8.4,8.4.1,8.4.2,8.5.1,8.5.2,8.6,8.7,9.1.9.1.3,10.2

01. Purpose:

To define methodology followed by Plasma Spray Processors for quality assurance.

02. Scope:

Produce and deliver product as per customer requirements

03. Process Owner:

DH QUALITY

04. Process Description:

Customer requirements are satisfied through continuous monitoring of all stages of Operation and till delivery of product.

05. Key Process Indicators:

- 01. On time delivery of product
- 02. Number of corrections required during and post Operation
- 03. Customer complaints

06. Abbreviations:

- ISO** International Organization for Standardization.
- PFD** Process Flow Diagram
- CAPA** Corrective action

07. Records:

Sl. No.	Record Title	Format No.	Retention Period	Retention Resp.	Indexing Method
1	List of measuring equipments	QF/QA/02	Till Rev.	HEAD QUALITY	Computer/Hard copy
2	Instrument & Gauge history card	QF/QA/03	Till Instrument Exists	HEAD QUALITY	Computer/Hard copy
3	Job card / In-process Inspection report	QF/QA/06	3 Year	HEAD QUALITY	Hard copy
4	Final Insp. Report	QF/QA/11	1 Year	HEAD QUALITY	Format/ Computer
5	Receipt Inspection Record	QF/QA/07	Till Rev.	HEAD QUALITY	Computer/Hard copy
6	Record of Revision Status	QF/QMS/05	Till Rev.	HEAD QUALITY	Hard Copy
7	Pre-Inspection Report	QF/QA/10A10B,10C	1 Year	HEAD QUALITY	Format/ Computer
Disposition: Shred/ Delete					
Prepared by		Reviewed by		Approved by	
Designation	DH QA	Designation	Partner	Designation	Partner
Date	01.04.2019	Date	01.04.2019	Date	01.04.2019

Plasma Spray Processors

Title: Quality assurance and calibration:

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ISO Clause: 6.2,7.1.5,7.1.5.1,7.1.5.2,8.2,8.2.1,8.2.2,8.4,8.4.1,8.4.2,8.5.1,8.5.2,8.6,8.7,9.1.9.1.3,10.2

Monitoring and Measurement Of Product And Processes

01. Purpose:

- To control the process of inspection and testing to ensure compliance with specified requirements

02. Scope:

This procedure applies to inspection and testing of purchased components, works completed by any repair at workshop

03. Process Owner:

DH QA with the help of QA Engineers is responsible for implementation of this procedure in their respective areas

04. Input:

Purchased Component and Component received for repairs, In-process & final components.

05. Procedure

5.1 Receiving Inspection

- QC Engineer will physically check purchased items against Purchase Order (PO) pertaining to component type, quantity, dimension and general condition.
- He shall reject damaged or nonconforming items at incoming stage and ensure the adequate storage and protection of received items.
- Inspection is done as per incoming component Inspection report QF/QA-07
- Records of receiving inspection of purchased items shall be maintained in Goods receipt note cum Incoming inspection cum supplier QF/ST-02.
- On receipt of Component for servicing repair, receiving inspection shall be carried out to assess condition of Component and to decide activities to be done during servicing / repairing

5.2 In-Process Inspection

- DH QA with the help of QA engineer shall continually monitor and inspect / test the semi-finished products **WILL GIVE APPROVAL FOR ONE COMPONENT IN A DAILY BASIS AND RELEASED FOR THE NEXT PROCESS (MASKING , BLASTING, SPRAYING, GRINDING, POISHING ETC).** according to specified requirements) . If a nonconforming condition arises, the "Nonconforming Product" label is raised by the concerned QA Engineer.
- For reworked items, they shall be re-inspected or repaired by Production supervisor
- All reworked items shall carry an appropriate identification tag.

5.3

None of the purchased items and/ or in-process components is passed on to next stage without inspection. Hence, positive recall system is not required

5.4 Final inspection and testing

- All repaired/ processed products are subjected to final inspection by the DH QA/QA/QC Engineer prior to delivery to the customer.
- Final inspection / test review includes customer inspection as applicable, to decide if finished products can be shipped to customers.
- Records of final inspection of repaired components are maintained in QF/QA/11
- Non-conformities observed during inspection / testing are handled as per procedure QMS-08.
- Inspection / Test Reports identify authority for product release.

6. Output and Process Effectiveness Monitoring

Inspected / tested incoming items, in process and finished rotors are outputs of this process. Effectiveness of process monitored on quarterly basis through technical customer complaints received

Prepared by		Reviewed by		Approved by	
Designation	DH QA	Designation	Partner	Designation	Partner
Date	30.11.2024	Date	30.11.2024	Date	30.11.2024

Plasma Spray Processors

Title: Quality assurance and calibration:

Ver. 00 / 01.04.2019 QMS-13

ISO Clause: 7.1.5, 7.1.5.1, 7.1.5.2

Monitoring and measuring resources

01. Purpose:

To ensure control of monitoring and measuring resources (inspection, measuring and test resources) so as to insure and confidence in measurements.

02. Scope:

- This system covers selection, calibration, handling and storage of inspection, measuring and test equipment at Plasma Spray Processors

03. Process Owner:

- DH QA

04. System

4.1 Selection of equipment

- Selection of monitoring and measuring equipment e.g. micrometers, vernier calipers, hardness tester, surface roughness meter, ultrasonic thickness tester, contact less thermometers, etc. is done according to measurement requirements

4.2 Care of equipment

- List of monitoring and measuring equipments which need calibration / verification is prepared.
- The list is updated on receipt of new measuring equipments and scrapping of defective measuring equipments.
- Measuring equipment shall be calibrated / verified periodically by approved external calibration agencies **or internal QA team**. Calibration certificates shall be received and verified.
- Calibration / verification is done with respect to reference standards traceable to Indian / international standards.
- Calibration status is identified by suitable tags / stickers on measuring equipments where feasible.
- Measuring equipments are handled, stored in such a way that they remain suitable for use, when required.

5 Defective Equipment

- When the measuring equipments is found defective, it is prevented from further use. The materials previously accepted / rejected by this defective measuring device are scrutinized for suitability of use by Plasma Spray Processors.

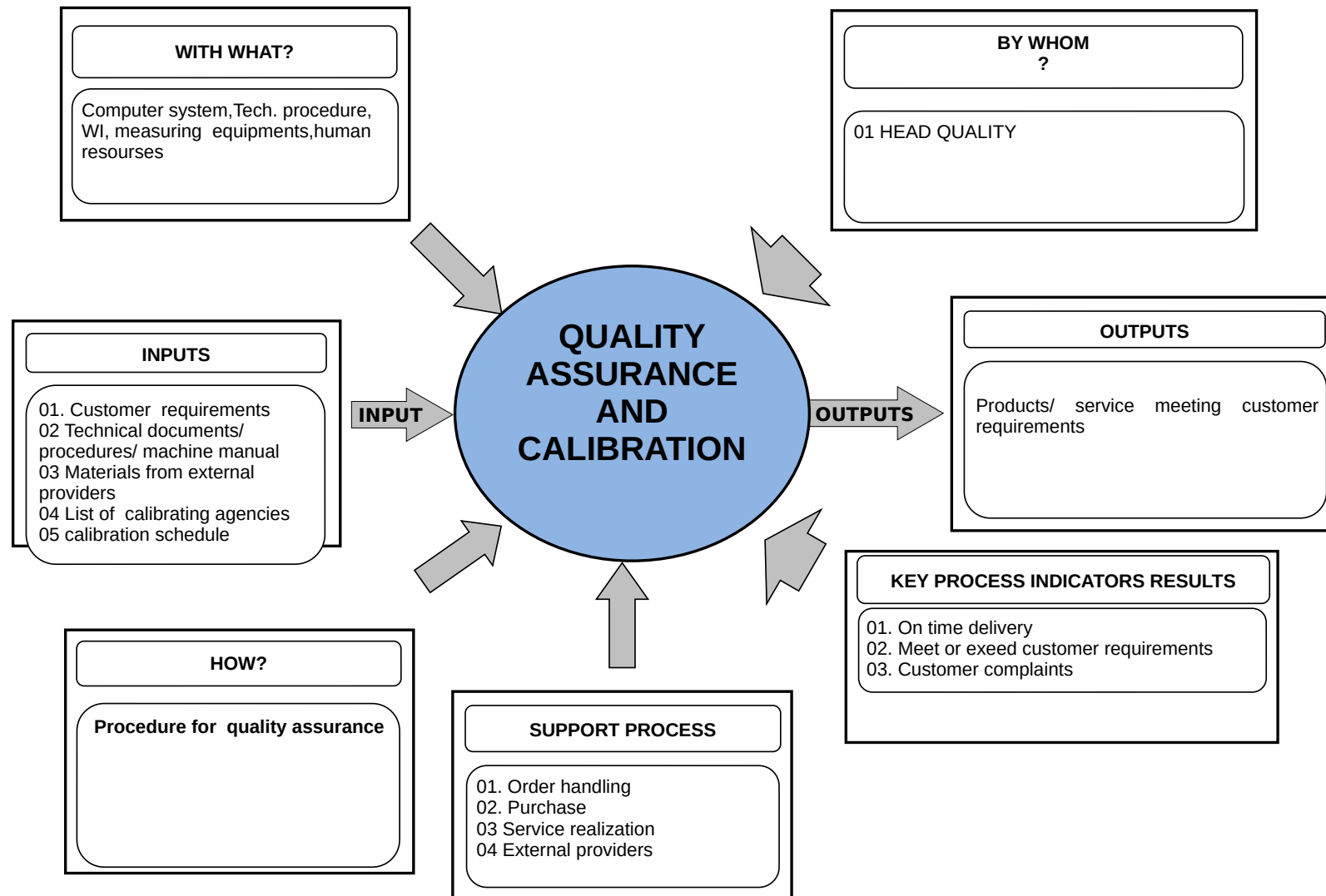
Prepared by		Reviewed by		Approved by	
Designation	DH QA	Designation	Partner	Designation	Partner
Date	01.04.2019	Date	01.04.2019	Date	01.04.2019

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Prepared by

Reviewed by

Approved by

partner

Designation	DH QA	Designation	Partner	Designation	Partner
Date	01.04.2019	Date	01.04.2019	Date	01.04.2019

Plasma Spray Processors

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ISO Clause:6.2,7.1.5,7.1.5.1,7.1.5.2,8.2,8.2.1,8.2.2,8.4,8.4.1,8.4.2,8.5.1,8.5.2,8.6,8.7,9.1.9.1.3,10.2

Source	Input	Description of Activity in Flow Chart	Out put	Risk considered	Control methods
Customer requirements on new products or /modification of current products	Customer order Machine manual	Planning for Quality	Checklist/ WI & Procedure	Wrong understanding	CFT review
	BOM OL/BO Parts	Inward inspection	Inward inspection report	Major crieteria not considered in QPD	Thorough review to avoid such missteps
	Process Operation Sub parts Subassemblies	In-process inspection of repair & processing process	Inprocess inspection report	Errors in processing not evidenced in inspection	Periodical updates of CL & Drawings
	Final product	Test product as per specification	Tested product	Misunderstanding of customer reqirements	Written communication from customer
	Tested product,	Handover to customer with final validation	Final inspection report		
	Measuring intruments Calibration schedule Approved agencies	Calibration of measuring instruments	Fit for use	Out of calibration instruments	Quarantine products and sort defective products
Prepared by		Reviewed by		Approved by	
Designation	DH QA	Designation	Partner	Designation	Partner
Date	01.04.2019	Date	01.04.2019	Date	01.04.2019

Plasma Spray Processors

Title: Quality assurance and calibration:

Ver. 00 / 09.05.2023 QMS-13

ISO Clause:

NDT VISUAL TESTING

1) SCOPE:

business
ethics...

2) REFERENCE DOCUMENTS

2.1 ASNT Recommended Practice SNT-TC-1A:
Personnel Qualification and Certification in Non-destructive Testing

3) QUALIFICATION OF TESTING

- 3.1 The inspection can be carried out by a NDT Level I or NDT Level II operator.
3.2 An operator with VT-Level 1 can perform an examination only under the supervision of personnel qualified and certified VT-Level 2 or higher.
3.3 Interpretation and evaluation of results shall only be carried out by personnel qualified and certified VT-Level 2 or higher, who is supervising the examination of VT-Level 1 and evaluates the results. Both have to sign the records.

4) TEST PROCEDURE

4.1 Examination procedure:

Direct visual examination may usually be made when access is sufficient to allow surface examination

- a) Within 600 mm (23.6") to the surface to be tested.
b) At an angle of not less than 30° to the surface to be tested.

4.2 Mirrors may be used to improve the angle of vision, aids such as a magnifying lens may be used to assist the examination.

4.3 For dimensional measurement, surface illumination shall be minimum 500 Lux.

4.4 The part under examination shall be illuminated with natural light or with artificial light source, necessary to attain a minimum of 1076 Lux, the light levels should be checked with a calibrated lux meter.

4.5 The lighting in the inspection area shall be such as to minimize any glare from the inspection surface.

5) ACCEPTANCE CRITERIA FOR MACHINED SURFACES

5.1 Relevant discontinuities

Relevant discontinuities are those that require consideration but may be acceptable. Any discontinuities that are less than 0.8 mm (1/32 inch) in length, width and depth shall not be considered relevant and can be ignored.

Cracks: A crack is defined as a visually apparent interruption to the continuity of the machined surface where the two sides of the interruption are close enough together to make it impossible to fully see the crack's bottom. If the bottom of the interruption is fully visible, then the discontinuity is a scratch, dent or gouge. While cracks are usually classified as linear discontinuities, they may be either linear or rounded. If visual examination presents doubt as to whether a visible mark is in fact an interruption to the surface (for example a line does not catch on one's fingernail) then other surface non-destructive methods should be used to determine the relevancy of the suspected discontinuity.

5.2 Linear and rounded discontinuities

5.2.1 The longest dimension of any indication shall be considered as the length 'L'.

5.2.2 The biggest dimension of the indication perpendicular to the length direction shall be considered as the Width 'W'

5.2.3 Indication with Length 'L' less than or equal to the width shall be considered to be a rounded indication:
 $L < 3 W$

5.2.4 Indication with Length 'L' more than the width shall be considered to be a linear indication: $L > 3 W$

5.3 ACCEPTANCE CRITERIA FOR GENERAL MACHINED SURFACES

5.3.1 General machined surfaces are any surfaces other than the following:

a) Thermal Spray Coating surfaces (See paragraph 5.3.1)

5.3.2 The following discontinuities are unacceptable for general machined surfaces:

- a) Cracks
- b) Linear discontinuities greater than 4.7 mm (3/16 inch).
- c) Rounded discontinuities or dents greater than 4.7 mm (3/16 inch) in length, width or depth or 1/3 the wall thickness in depth whichever is less.
- d) Any group of 4 or more relevant discontinuities on a given surface of 2,580 mm² (4 square inches) in area.
- e) Adjacent relevant discontinuities with less than 25 mm (1 inch) separation.

5.4 COATED SURFACES

5.4.1 The following discontinuities are unacceptable on thermal spray coating surfaces.

- a) Cracks
- b) Rounded or linear discontinuities, dents or scratches greater than 1.5 mm (0.06 inch) in length, width or depth.
- c) Any group of 4 or more relevant discontinuities on a given surface of 6,450 mm² (10 square inches) in area.
- d) Adjacent relevant discontinuities with less than 25 mm (1 inch) separation.

6) All surfaces of received components shall be inspected to detect any abnormalities such as :

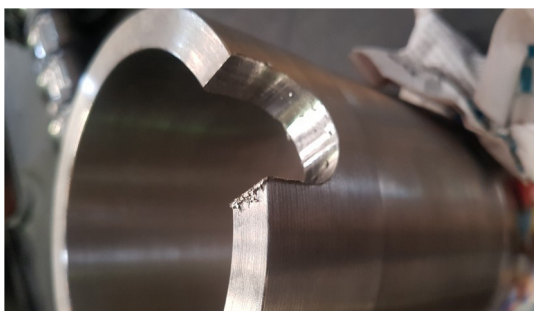
AREAS I	AREAS II	AREAS III
Cracks	Cracks	Cracks
Dents / Scratches	Pin holes	Dent / Scratches
Corrosion pitting	Porosities	Cobweb - Spider webs
Grooving	Oil oozing	Chip offs
Arc strikes	Pitting	Overspray
Scoring		Oil oozing
Chipping		Pin holes
Presence of Burrs		Voids
Tool marks		Delamination
Coolant inside bore		Contamination

Areas 'I': **SURFACE INSPECTION OF AS RECEIVED SURFACES** shall be marked on the job, photographed, measured and a report sent to the supplier for his comments or to be repaired with supplier's approval.

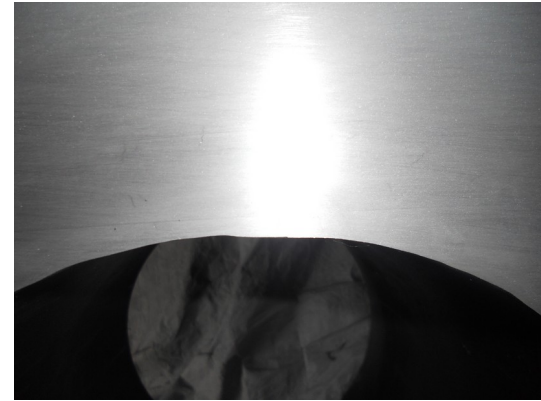
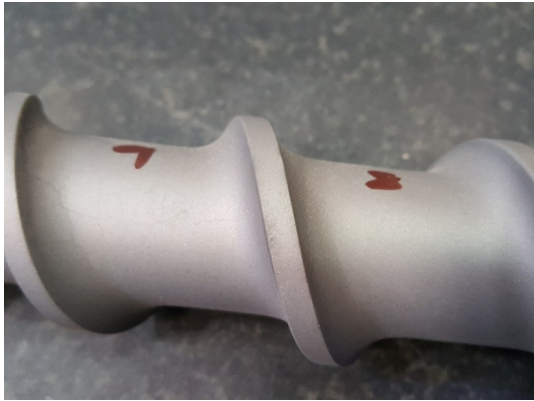
Areas 'II': **DISCONTINUITIES FOR GRIT BLASTED SURFACES** shall be marked on the job, photographed, measured and a report sent to the supplier for his comments or to be repaired with supplier's approval. But in any circumstances the discontinuities should not be waived off.

Area 'III': **DISCONTINUITIES IN SPRAY COATED SURFACES** related to cracks, dents, scratches, Cob-web, chip off, is to recoat the components. Other discontinuities can be cleaned, polished, machined, ground etc.

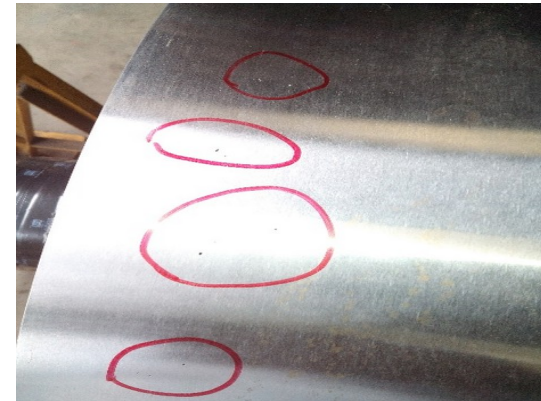
BURRS



CRACKS

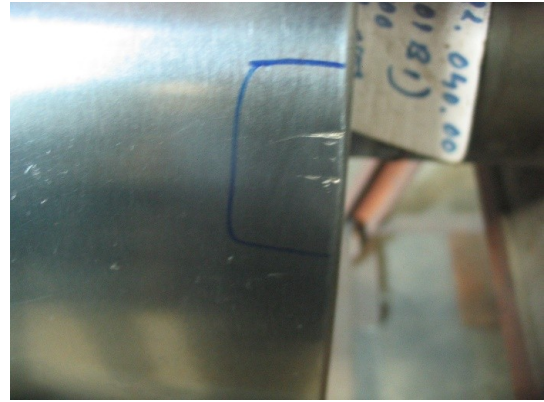


PIN HOLES



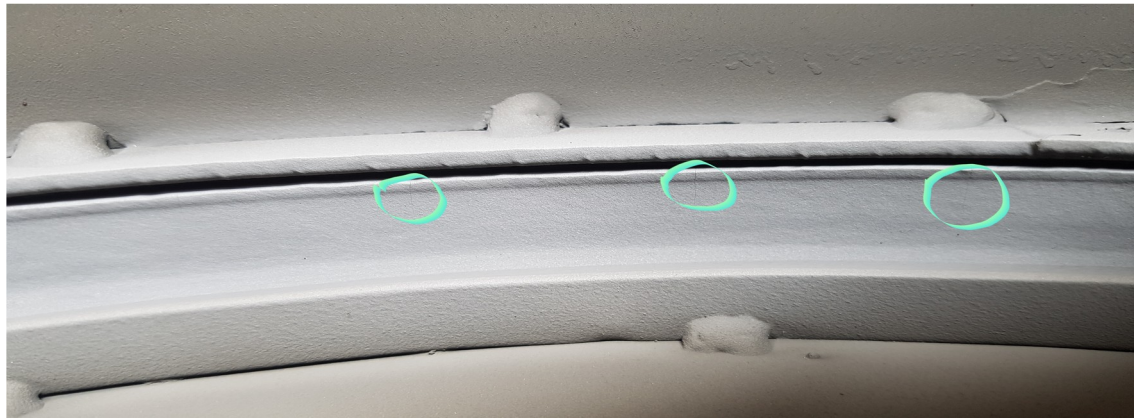


DENTS



PIN HOLES & CRACKS VISIBLE AFTER COATING





7)REPORTING

For each official visual examination required in the Project Quality Plan (Inspection & Test Plan) for NDE purposes and for each item tested, there shall be a completed a report. This report has to include at least following information:

- Job and part identification.
- Surface condition.
- Test procedure reference and method used.
- Equipment used.
- Results (of the recording indications: size, depth, position and classification of defects, eventually with drawing).
- The name, qualification (Level) and the signature of the person(s) carrying out the test respectively having made the interpretation and evaluation of indications.

**** Components shall be clearly identified as being either acceptable or rejectable.**

Acceptable Components to place at respective area and Rejectable Components (Small Components) to be kept in Quarantine Area/ and Large Components Job Cards to be kept in Quarantine Area.

****Non-conforming areas on components shall be marked on the components themselves, or where this is not possible on an NDE defect Map.**

Circle the defective area with Permanent Marker.

Prepared by		Reviewed by		Approved by (ASNT LEVEL III)	
Designation	DH QA	Designation	Partner		
Date	09.05.2023	Date	09.05.2023	Date	09.05.2023

PLASMA SPRAY PROCESSORS

QUALITY CONTROL PLAN (QCP)											
PURCHASE ORDER NO				DATE		CTQ: Critical to Quality			PSP/QCP/ 01 /00 12.10.2023 QMS-13		
CTP Critical to Process,											
SR. NO.	DESCRIPTION	CHARACTERISTICS VERIFIED	MACHINE/MANUAL	METHODS OF CHECK	QUANTUM OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORM	FORMAT OF RECORD	PSP/ CUSTOMER	REVIEW/ PERFORM/ OBSERVATION	REACTION PLAN
1	Incoming Component Inspection	Surface Condition	N/A	Visual	100%	QF/QMS/02/12	CTQ	Pre-Inspection Report	PSP	QC	Inform customer for approval & tag.
		Dimensional Inspection		Micrometer	100%		CTQ				
		De-Coating/Pre-Grinding		Visual	100%		CTP				
2	Cleaning	Surface Area	Manual	Visual	100%	QF/QMS/02/15	CTP	Job Card	PSP	QC	Oil/ Grease/Burr traces found, Redo it untill it is completely clean.
		Rusting & Burr	Emery Paper / Pneumatic Tool	Visual	100%		CTP				
3	Removal of Old Coating & Pre Machining/ Pre-Grinding	Peeled Off Coating	Grit Blasting M/c. Manually	Visual	100%	QF/QMS/02/02	CTP	Job Card	PSP	QC	Incase coating found re-blast
		Surface Dents/ Uneven Wear	Lathe M/c./ Grinding M/c	Micrometer	100%						Machine till the surface is Uniform
4	Test	Dye Penetrant Test	Penetrant /	Visual	100%	QF/QMS/02/26	CTQ	Report	Testing Lab	QC/THIRD PARTY	If Failed, Report to Customer
		Ultra sonic Test		Visual	100%		CTQ	Report	Testing Lab		
		Magnectic Particle Test		Visual	100%		CTQ	Report	Testing Lab		
5	Surface Preparation	Masking	Manual	Visual	100%	QF-QMS/02/04	CTP	Job Card	PSP	QC	If the surface is not uniformly white, reblast the surface
		Grit Size & Pressure	Grit Blasting M/c.	Visual	100%						
		Surface Roughness	Surface Roughness Tester	Comparter / Ra Value	2 to 3 Location						
		Cleaning	Air	Visual	100%						
6	Spraying	Selection of Process	Plasma/HVOF/Twin Wire Arc/Flame Spray	Visual	100%	QF/QMS/02/06 QF/QMS/02/07 QF/QMS/02/08	CTQ	Job Card	PSP	QC	N/A
		Powder Parameter	Machine/Manual	Visual							
		Job Speed	Lathe M/c. / Turn Table	VFD Display							
		GunSpeed	Manipulator/ Robot	VFD/ Robot Program							
		Pre-heating	Infra red Thermometer	40° to 80°C							
		Powder/ Wire Spraying	Gun	Thickness/pass, Micrometer & Electro Magnetic Probe	Intermediate						If overheated/peeled off/over size respray (to follow from process 3). A NC to be reported.
7	Cooling of Component	N/A	Infra red Thermometer	Room Temperature	N/A	N/A	N/A	N/A	PSP		N/A
8	Finishing	Dimensions and Run out	Lathe M/c, Grinding M/c	Micrometer & Dial Guage	2 to 3 Location/ each	QF/QMS/02/13	CTQ	Job Card	PSP	QC	If NC tag the component with Red Tag & Respray
		Surface Finish	Surface Roughness Tester	Ra Vaule							
9	Final Inspection	Dimensional Inspection	Instrument	Micrometer	100%	QF/QMS/02/10	CTQ	Final Inspection Report	PSP / Customer	QC	If NC tag the component with Red Tag & Respray
		Surface Roughness	Surface Roughness Tester	Ra Vaule							
10	Sealant	Microporosity	Manual	Visual	100%	N/A	N/A	Job Card	PSP	QC	N/A
11	Packing	To protect coated surface	Manual	Visual	100%	QF/QMS/02/25	N/A	Job Card	PSP	QC	N/A
12	Dispatch	Quantity, PO, OGP, Inspection Report	Software	Visual	100%	QF/QMS/02/11	N/A	Job Card	PSP	ADMIN	N/A